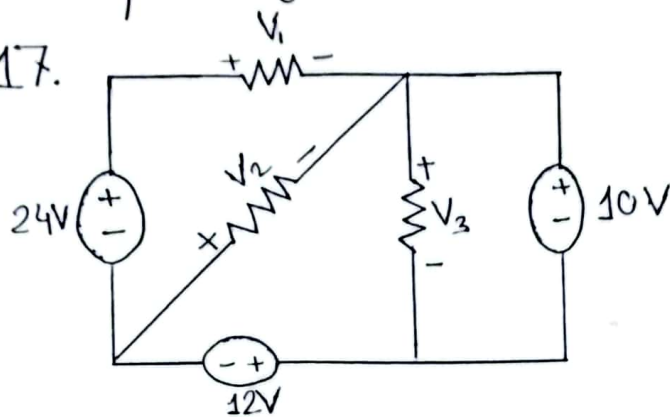


Chapter-02:

17.



Obtain V_1 through V_3 in the circuit.

Solution:

Applying KVL around the entire outside loop,
we get:

$$-24 + V_1 + 10 + 12 = 0$$

$$\Rightarrow V_1 = 24 - 10 - 12$$

$$\therefore V_1 = 2 \text{ volt.}$$

Applying KVL around the loop containing V_2 ,
the 10V source and the 12V source, we get:

$$V_2 + 10 + 12 = 0$$

$$\Rightarrow V_2 = -10 - 12$$

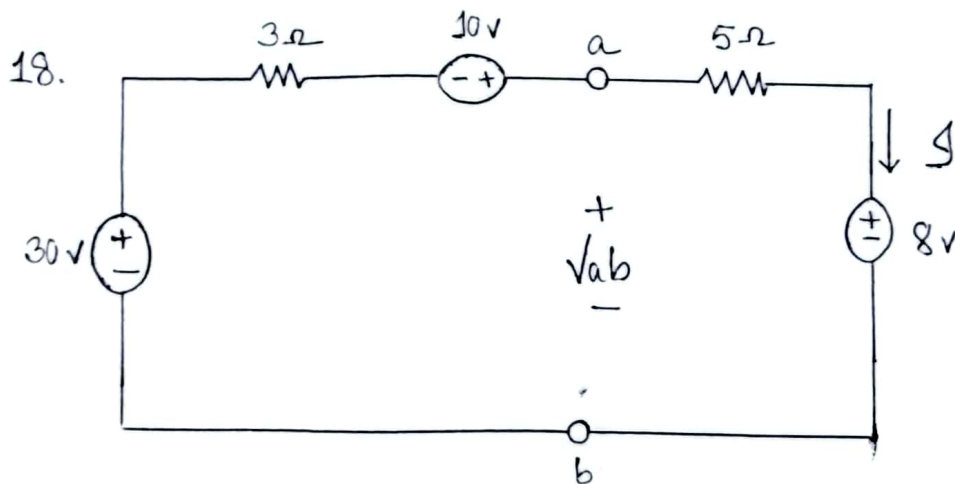
$$\therefore V_2 = -22 \text{ volt.}$$

Applying KVL around the loop containing v_3 and 10V source, we get:

$$-v_3 + 10 = 0$$

$$\Rightarrow v_3 = 10 \text{ volt.}$$

Ans: $v_1 = 2 \text{ volt}$
 $v_2 = -22 \text{ volt.}$
 $v_3 = 10 \text{ volt.}$



Find I and v_{ab} in the circuit.

Solution: Applying KVL in the circuit, we get:

$$-30 + 3I - 10 + 5I + 8 = 0$$

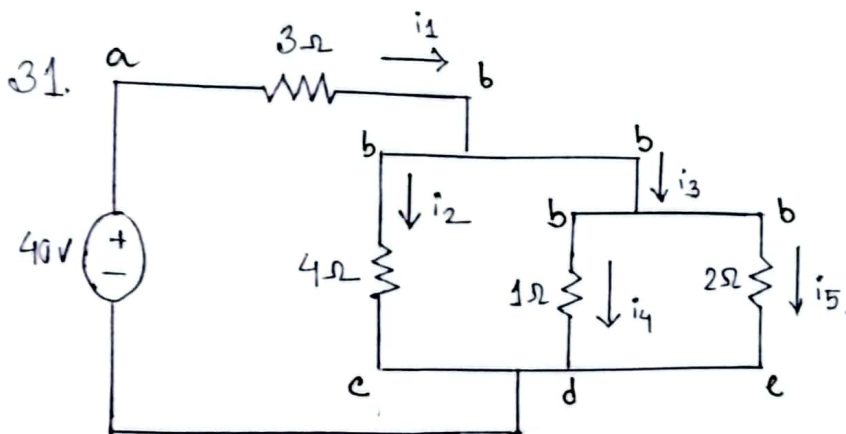
$$\Rightarrow 8I = 32$$

$$\therefore I = 4 \text{ A.}$$

If we were to connect a and b, considering the loop containing V_{ab} source, 5Ω resistor and $8V$ source, applying KVL, we get:

$$\begin{aligned} -V_{ab} + 5I + 8 &= 0 \\ \Rightarrow -V_{ab} + 5(4) + 8 &= 0 \\ \therefore V_{ab} &= 28 \text{ volt.} \end{aligned}$$

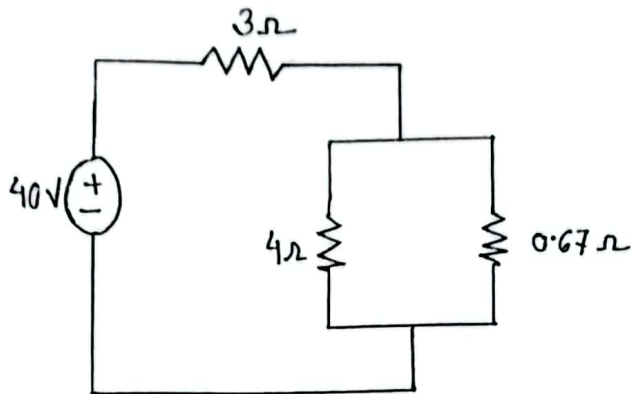
Ans: $V_{ab} = 28 \text{ volt.}$, $I = 4 \text{ A.}$



For the circuit, determine i_1 to i_5 .

Solution: In the circuit, we notice that 1Ω and 2Ω are in parallel connection.

$$\begin{aligned} \therefore \frac{1}{R_p} &= \frac{1}{1} + \frac{1}{2} \\ \Rightarrow R_p &= \left(\frac{3}{2}\right)^{-1} \\ \therefore R_p &= 0.67 \Omega \end{aligned}$$

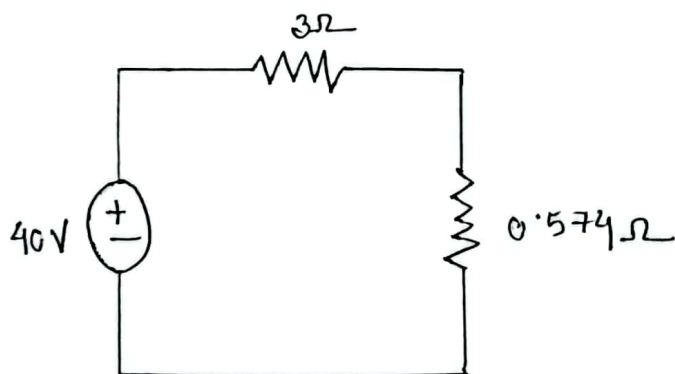


Note, 4Ω and 0.67Ω are also in parallel connection with each other.

$$\therefore \frac{1}{R_{p'}} = \frac{1}{4} + \frac{1}{0.67}$$

$$\Rightarrow R_{p'} = (1.7425)^{-1}$$

$$\therefore R_{p'} = 0.574\Omega$$



Finally, 3Ω and 0.574Ω are in series connection.

$$\begin{aligned} \therefore R_{eq} &= 3 + 0.574 \\ &= 3.574\Omega \end{aligned}$$

Total current in the circuit:

$$i_1 = \frac{40}{3.5714} = 11.2 \text{ A.}$$

$$\begin{aligned} \text{and, } V_{p'} &= i_1 R_{p'} \\ &= 11.2 \times 0.5714 \\ &= 6.39 \text{ V} \end{aligned}$$

$$\therefore i_2 = \frac{6.39}{4} = 1.59 \text{ A.}$$

again, $\therefore i_3 = i_1 - i_2 = 11.2 - 1.59 = 9.61 \text{ A.}$

$$\begin{aligned} V_p &= i_3 R_p \\ &= 9.61 \times 0.67 \\ &= 6.409 \text{ V} \end{aligned}$$

$$\therefore i_4 = \frac{6.409}{1} = 6.409 \text{ A.}$$

$$i_5 = \frac{6.409}{2} = 3.205 \text{ A.}$$

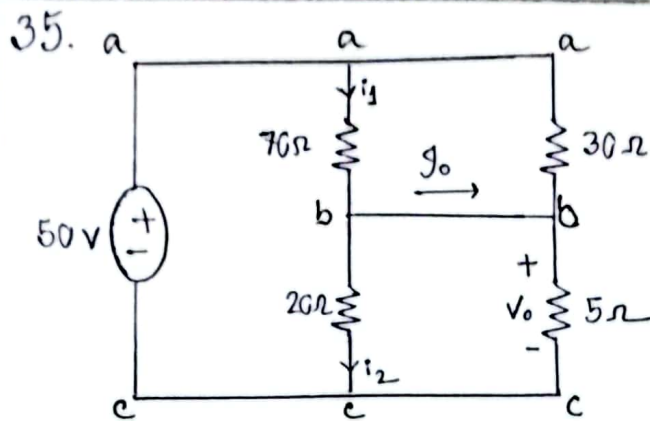
Ans: $i_1 = 11.2 \text{ A.}$

$$i_2 = 1.59 \text{ A}$$

$$i_3 = 9.61 \text{ A}$$

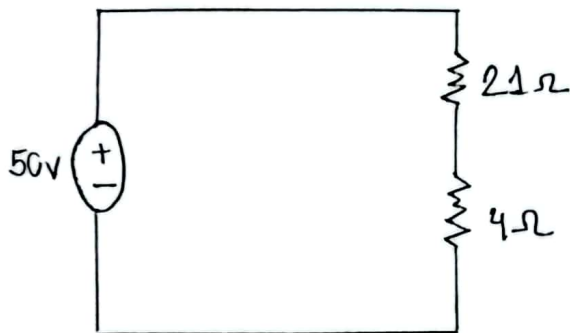
$$i_4 = 6.409 \text{ A}$$

$$i_5 = 3.205 \text{ A.}$$



Calculate V_o and I_o in the circuit.

Solution:



From the circuit, we note that 70Ω and 30Ω are in parallel connection and so are 20Ω and 5Ω respectively.

$$\therefore R_{p1} = \left(\frac{1}{70} + \frac{1}{30} \right)^{-1} \quad \text{and} \quad R_{p2} = \left(\frac{1}{20} + \frac{1}{5} \right)^{-1}$$

$$= 21\Omega \quad \quad \quad = 4\Omega$$

Then, 21Ω and 4Ω are in series connection.

$$\therefore R_{eq} = 21 + 4 = 25\Omega$$

$\therefore$ Total current flow in the circuit,

$$I = \frac{50}{25} = 2A.$$

$$\text{Now, } V_{p2} = I R_{p2} \quad \text{and } V_{p1} = I R_{p1}$$

$$= 2 \times 4 \quad = 2 \times 21$$

$$= 8 \text{ volt.} \quad = 42 \text{ volt}$$

$$i_1 = \frac{42}{70} = 0.6A.$$

$$\text{and } i_2 = \frac{8}{20} = 0.4A.$$

Using KCL at node b:

$$i_1 = i_2 + I_0$$

$$\Rightarrow 0.6 = 0.4 + I_0$$

$$\Rightarrow I_0 = 0.6 - 0.4$$

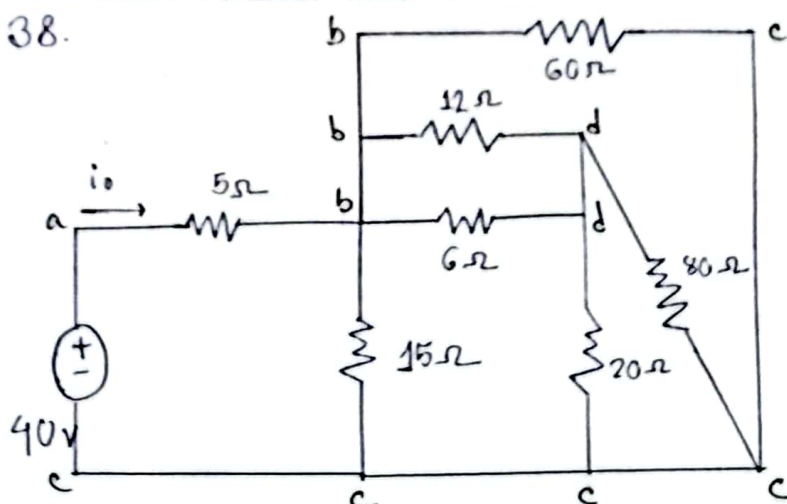
$$\therefore I_0 = 0.2A.$$

Ans: $I_0 = 0.2A$

$$V_0 = V_{p2} = 8 \text{ volt.}$$

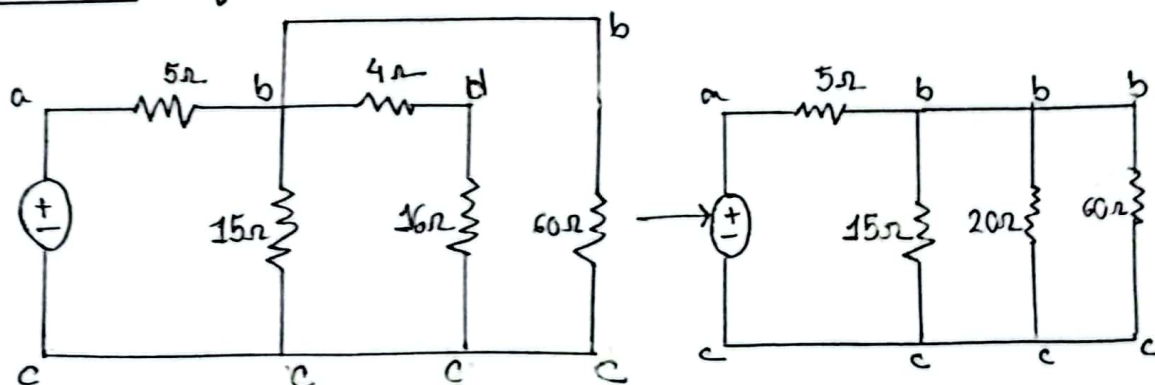


38.



Find Req and go in the circuit

Solution: If the circuit is redrawn:



$12\Omega, 6\Omega$ and $80\Omega, 20\Omega$ are both in parallel connection, we can write:

$$R_{p1} = \left(\frac{1}{12} + \frac{1}{6} \right)^{-1} = 4\Omega$$

$$R_{p2} = \left(\frac{1}{20} + \frac{1}{80} \right)^{-1} = 16\Omega$$

Now, R_{p1} and R_{p2} are in series connection

$$\begin{aligned} \therefore R_s &= R_{p1} + R_{p2} \\ &= 4 + 16 \\ &= 20\Omega \end{aligned}$$

Now, 15Ω , 20Ω and 60Ω are in parallel connection to each other.

$$\therefore R_p = \left(\frac{1}{15} + \frac{1}{20} + \frac{1}{60} \right)^{-1} = 7.5\Omega$$

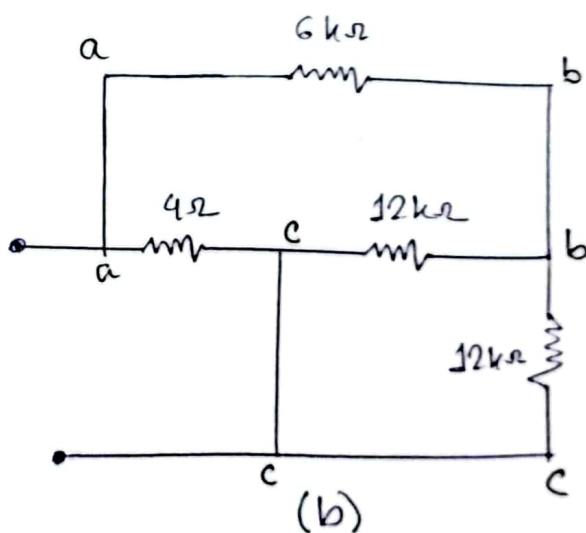
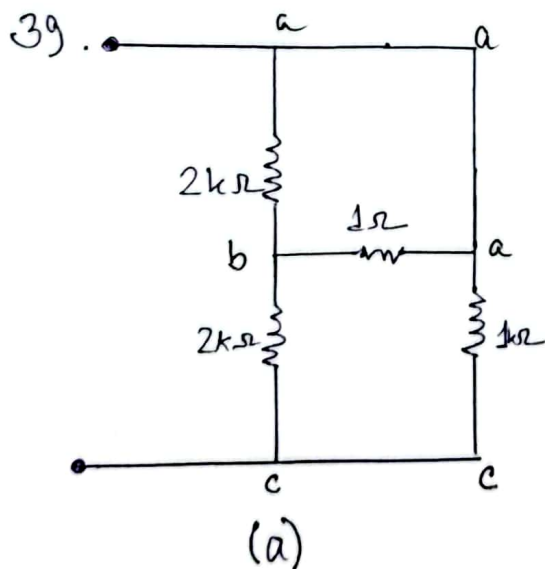
Finally 7.5Ω is connected in series with 5Ω

$$\therefore R_{eq} = 5 + 7.5 = 12.5\Omega$$

$\therefore$ Total current in the circuit,

$$I_0 = \frac{40}{12.5} = 3.2\text{ A.}$$

Ans: $I_0 = 3.2\text{ A}$, $R_{eq} = 12.5\Omega$



Evaluate R_{eq} for each of the circuits.

Solution:

(a) From a and b node, $2k\Omega$ and $1k\Omega$ are in parallel connection to each other.

$$\therefore R_p = \left(\frac{1}{1} + \frac{1}{2} \right)^{-1} = 0.667 k\Omega$$

Again R_p is connected in series with $2k\Omega$.

$$\therefore \text{we get: } R_s = (0.667 + 2) k\Omega = 2.667 k\Omega$$

Finally R_s is connected parallelly with $1k\Omega$

$$\therefore R_{eq} = \left(\frac{1}{1} + \frac{1}{2.667} \right)^{-1} = 0.727 k\Omega$$

Ans

(b) From the figure:

Two $12k\Omega$ resistances are in parallel connection.

$$\therefore R_p = \left(\frac{1}{12} + \frac{1}{12} \right)^{-1} = 6 k\Omega$$

R_p is connected in series with $6k\Omega$

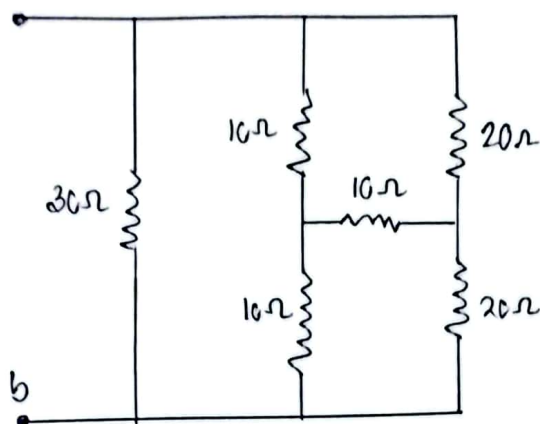
$$\therefore R_s = (6 + 6) = 12 k\Omega$$

R_s is connected parallelly with $4k\Omega$

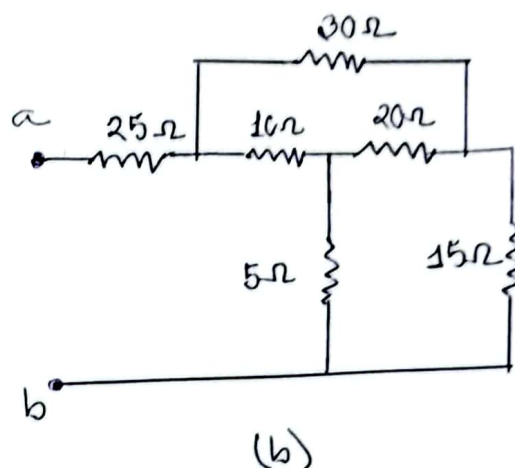
$$\therefore R_{eq} = \left(\frac{1}{12} + \frac{1}{4} \right)^{-1} = 3 k\Omega$$

Ans.

51. a



(a)

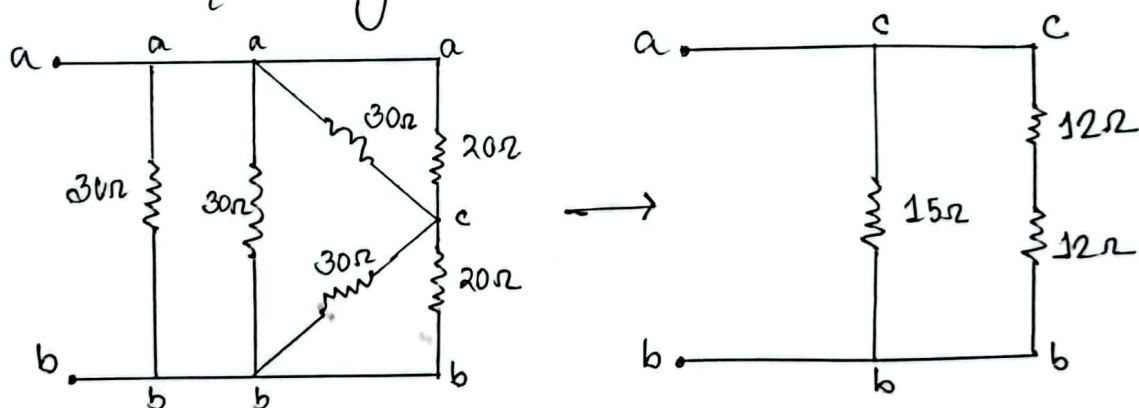


(b)

Obtain equivalent resistances at terminals a-b.

Solution:

(a) Converting the T-subnetwork into an equivalent Δ -network gives:



$$R_1 = \frac{10 \times 10 + 10 \times 10 + 10 \times 10}{10} = 30\Omega$$

Similarly, R_2 and R_3 are also 30Ω

Now, in nodes a-c and c-b, 30Ω and 20Ω are connected parallelly twice.

We can write:

$$R_{p1} = \left(\frac{1}{30} + \frac{1}{20} \right)^{-1} = 12 \Omega$$

$$R_{p2} = \left(\frac{1}{30} + \frac{1}{20} \right)^{-1} = 12 \Omega$$

R_{p1} and R_{p2} are connected in series.

$$\therefore R_s = 12 + 12 = 24 \Omega$$

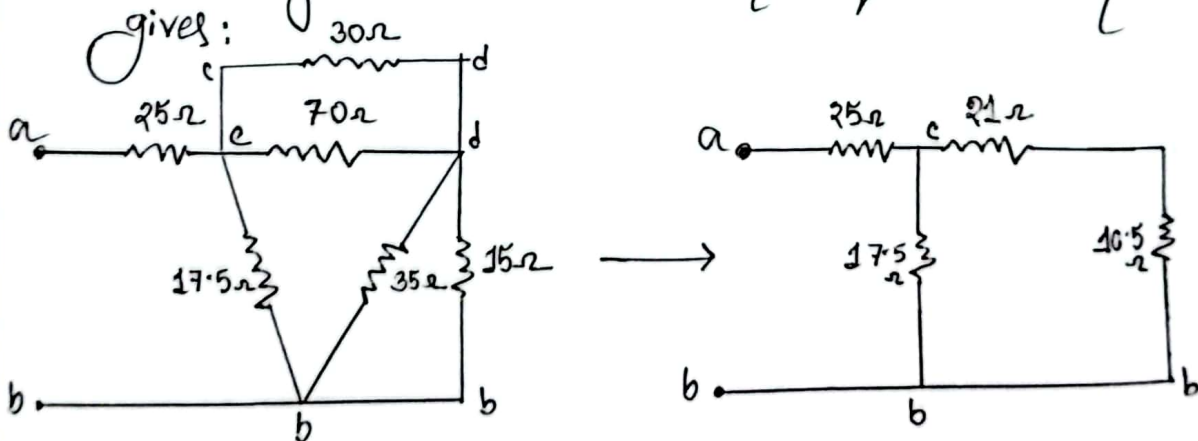
Again, two 30Ω are connected in parallel connection

$$\therefore R_{p3} = \left(\frac{1}{30} + \frac{1}{30} \right)^{-1} = 15 \Omega$$

Finally R_s and R_{p3} are in parallel connection

$$\therefore R_{eq} = \left(\frac{1}{15} + \frac{1}{24} \right)^{-1} = 9.231 \Omega \quad \text{Ans}$$

(b) Converting the T-subnetwork into equivalent Δ -network gives:



$$R_1 = \frac{10 \times 20 + 20 \times 5 + 10 \times 5}{10} = 35 \Omega$$

$$R_2 = \frac{10 \times 20 + 20 \times 5 + 10 \times 5}{20} = 17.5 \Omega$$

$$R_3 = \frac{10 \times 20 + 20 \times 5 + 10 \times 5}{5} = 70 \Omega$$

Now, 35Ω and 15Ω are connected parallelly.

$$\therefore R_p = \left(\frac{1}{35} + \frac{1}{15} \right)^{-1} = 10.5 \Omega$$

And, 30Ω and 70Ω are connected parallelly

$$\therefore R_{p'} = \left(\frac{1}{30} + \frac{1}{70} \right)^{-1} = 21 \Omega$$

R_p and $R_{p'}$ are in series connection.

$$\therefore R_s = 10.5 + 21 = 31.5 \Omega$$

Again, R_s and 17.5Ω are in parallel connection.

$$\therefore R_{p''} = \left(\frac{1}{31.5} + \frac{1}{17.5} \right)^{-1} = 11.25 \Omega$$

Finally, 25Ω and $R_{p''}$ are in series connection

$$\begin{aligned} \therefore R_{eq} &= 11.25 + 25 \\ &= 36.25 \Omega \end{aligned}$$

Ans.